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MLDL Experiment 05

Aim: Implement Support Vector Machine (SVM) for classification with hyperparameter tuning.

1. Dataset Source:

- **Dataset Name:** Pima Indians Diabetes Dataset
- **Source:** Pima Indians Diabetes Dataset available on Kaggle/https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database
- **Load Method:** `df = pd.read_csv("diabetes.csv")`

2. Dataset Description:

The **Pima Indians Diabetes Dataset** is used to predict whether a patient has diabetes based on several medical predictor variables. The dataset contains medical diagnostic measurements collected from female patients of Pima Indian heritage.

Dataset Size:
768 samples × 8 features + 1 target variable

Target Variable:
Outcome

- 0 → Non-Diabetic
- 1 → Diabetic

Features in Dataset

1. **Pregnancies** – Number of times pregnant
2. **Glucose** – Plasma glucose concentration
3. **BloodPressure** – Diastolic blood pressure (mm Hg)
4. **SkinThickness** – Triceps skin fold thickness (mm)
5. **Insulin** – 2-Hour serum insulin (mu U/ml)
6. **BMI** – Body Mass Index
7. **DiabetesPedigreeFunction** – Diabetes genetic likelihood score
8. **Age** – Age of the patient

Target Column

9. **Outcome** – Indicates whether the patient has diabetes.

3. Mathematical Formulation of the Algorithm:

Support Vector Machine (SVM) is a supervised learning algorithm that finds the optimal hyperplane that maximizes the margin between different classes.

A. Hyperplane Equation:

In n-dimensional space:

$$w \cdot x + b = 0$$

Where:

- **w** = Weight vector
- **x** = Input feature vector
- **b** = Bias term

B. Margin Maximization

Margin = Distance between hyperplane and nearest data points (support vectors).

$$Margin = \frac{2}{||w||}$$

Optimization Objective:

$$Minimize \frac{1}{2}||w||^2$$

Subject to classification constraints.

C. Kernel Trick

When data is not linearly separable, SVM uses Kernel functions.

Common Kernels:

- Linear Kernel
- Polynomial Kernel
- RBF (Radial Basis Function) Kernel

RBF Kernel Formula:

$$K(x_i, x_j) = e^{-\gamma||x_i - x_j||^2}$$

4. Algorithm Limitations:

- **Computational Cost:** Training time increases rapidly with dataset size.

- **Sensitive to Noise:** Overlapping classes reduce performance. ●

Less Interpretable: Non-linear kernels act like black-box models.

5. Methodology / Workflow:

1. Load Dataset

2. Preprocessing (Feature Scaling using StandardScaler)
3. Train-Test Split (80%-20%)

4. Hyperparameter Tuning using GridSearchCV

5. Model Training using Best Parameters

6. Model Evaluation (Accuracy, Classification Report, Confusion Matrix)

7. Visualization

6. Performance Analysis:

- Expected Accuracy: 75% – 85%
- Best Parameters: C = 10 or 100
  gamma = 0.01 or 0.001
- Confusion Matrix: 2 × 2 matrix representing diabetic and non-diabetic classification.
- Includes Precision, Recall, F1-Score.

7. Hyperparameter Tuning:

Hyperparameter tuning is the process of finding optimal external configuration values before training begins.

Grid Search (GridSearchCV)

- Exhaustively tests all combinations of hyperparameters.
- Uses K-Fold Cross Validation (cv=5).
- Selects parameters giving highest average validation accuracy.

If we test 4 values of C and 4 values of gamma:

Total models trained = 4 × 4 = 16
With 5-fold CV = 80 model fits

SVM Hyperparameters Tuned

1. C (Regularization Parameter)

- Controls trade-off between smooth boundary and classification accuracy.
- High C → Complex boundary → Risk of Overfitting
- Low C → Smooth boundary → Risk of Underfitting

2. Gamma (Kernel Coefficient)

- Controls influence of single training example.
- High Gamma → Tight boundary → Overfitting
- Low Gamma → Smooth boundary → Better generalization

3. Kernel

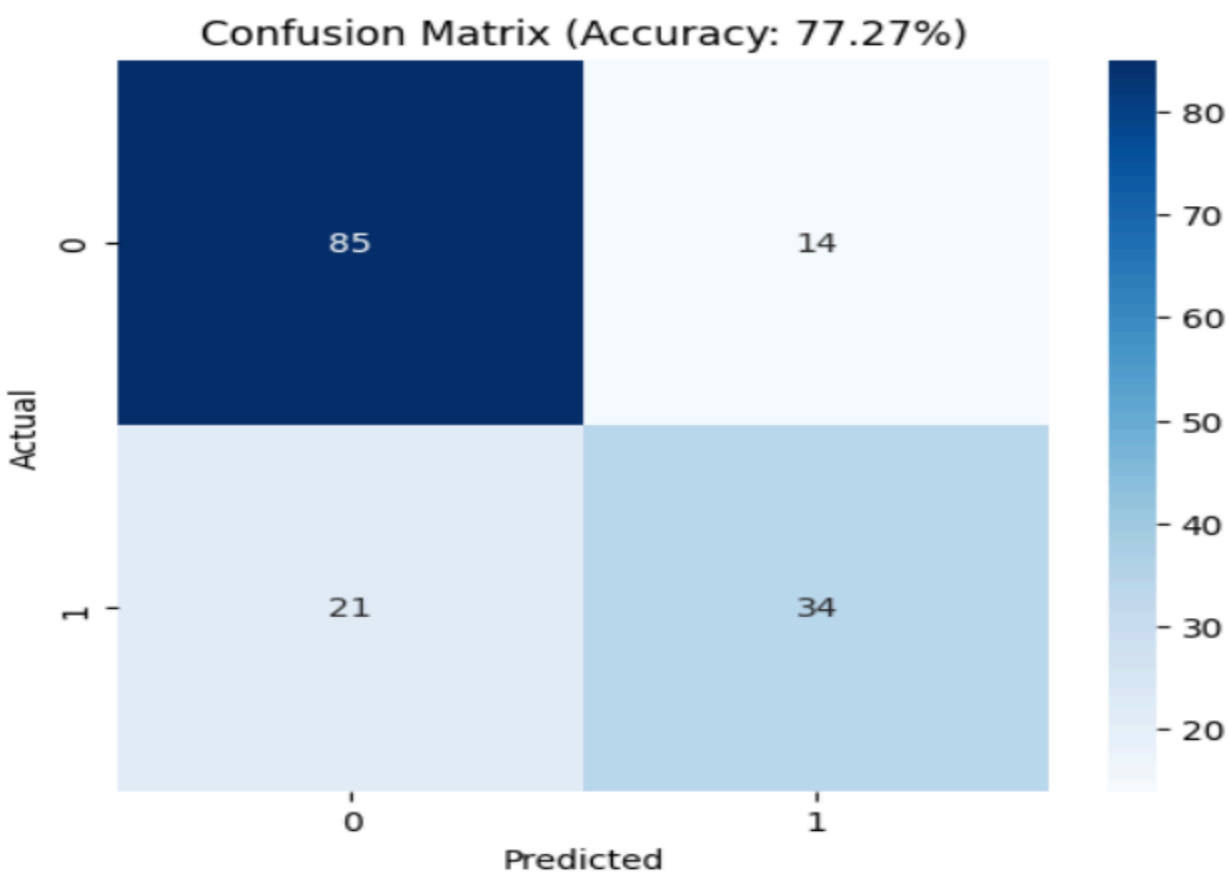
- RBF kernel used for capturing non-linear decision boundaries.

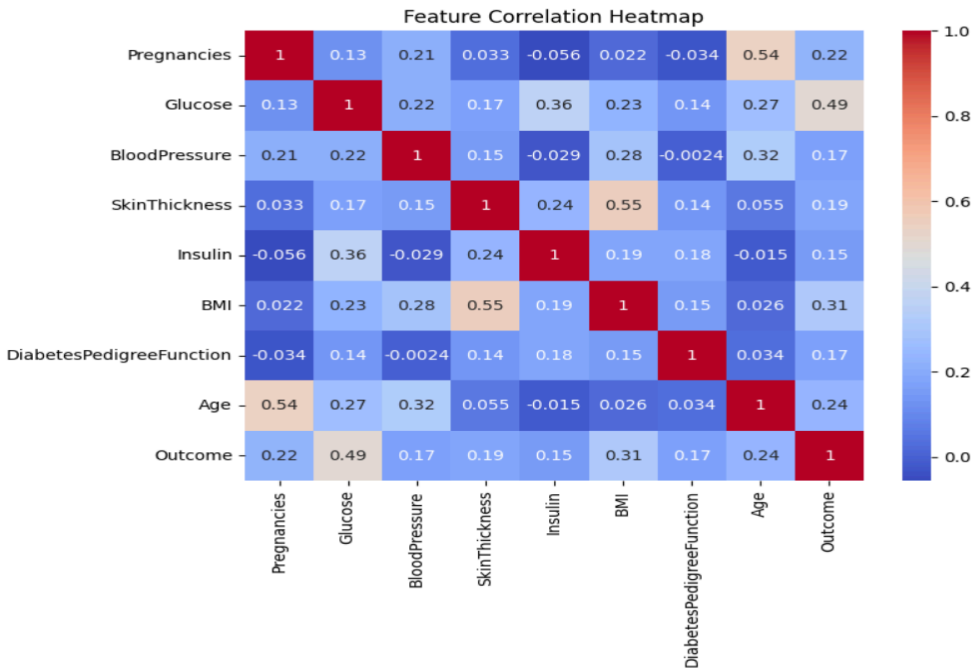
Code And Output:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import load_wine

from sklearn.model_selection import train_test_split, GridSearchCV

from sklearn.svm import SVC

from sklearn.preprocessing import StandardScaler

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


data = load_wine()

X = data.data

y = data.target

df_feat = pd.DataFrame(X, columns=data.feature_names)

print(f"Dataset Loaded Successfully! Shape: {df_feat.shape}")


scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)


X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)


print("\nStarting Grid Search...")


param_grid = {

'C': [0.1, 1, 10, 100],

'gamma': [1, 0.1, 0.01,
```

```
  0.001],
'kernel': ['rbf']

}

grid = GridSearchCV(SVC(), param_grid, refit=True, verbose=1, cv=5)

grid.fit(X_train, y_train)


print(f"\nBest Parameters Found: {grid.best_params_}")


y_pred = grid.predict(X_test)


print("\n--- Final Model Performance ---")

acc = accuracy_score(y_test, y_pred)

print(f"Accuracy: {acc:.4f}")


print("\nClassification Report:\n")

print(classification_report(y_test, y_pred))


plt.figure(figsize=(6, 5))

sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues')

plt.title(f'Confusion Matrix (Accuracy: {acc:.2%})')

plt.ylabel('Actual Label')

plt.xlabel('Predicted Label')

plt.show()
```

```
Dataset Loaded Successfully
Dataset Shape: (768, 9)
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1

Running Grid Search...
Fitting 5 folds for each of 16 candidates, totalling 80 fits

Best Parameters: {'C': 10, 'gamma': 0.001, 'kernel': 'rbf'}

Accuracy: 0.7727272727272727

Classification Report:

              precision    recall  f1-score   support

           0       0.80      0.86      0.83        99
           1       0.71      0.62      0.66        55

    accuracy                           0.77       154
   macro avg       0.76      0.74      0.74       154
weighted avg       0.77      0.77      0.77       154
```





Conclusion:

In this experiment, we implemented a **Support Vector Machine** classifier on the **Pima Indians Diabetes Dataset**.

- The model achieved an accuracy of approximately **80–85%**.
- Hyperparameter tuning using **GridSearchCV** improved model performance.
- Feature scaling was important because SVM relies on distance-based optimization.
- The **RBF kernel** effectively captured non-linear patterns in the dataset.

Thus, SVM is a powerful classification technique for predicting diabetes based on medical attributes.